

Force and laws of motion: Construction

Concept Book

Newton's first law and free-body diagrams

Learning promise

Grade-9 Force and Laws of Motion plus the owner-approved bounded Pinnacle preparation slice: balanced/unbalanced force reasoning, body-specific FBD ownership, friction direction, static-friction limits and kinetic-friction model, $F_{\text{net}} = m a$, Newton-third-law interaction pairs, common-acceleration connected bodies, ideal-string tension, and one fixed-pulley single-string length constraint. Observer-frame/pseudo-force reasoning remains an explicit-demand extension.

Concept route

1. Friction opposes relative sliding at a contact

1. Before giving friction a direction, imagine the contact made perfectly slippery and ask which way the surfaces would slide relative to each other; friction opposes that, and is zero if there is no such tendency.
2. Quantitative static and kinetic friction conditions
 1. Find the actual normal force and the friction that rest would need, compare that need with $\mu_s N$, and use μN as the friction value only at impending slip or once sliding.

Foundation route

- One-dimensional motion — direct route unavailable in this product

Model contract

- Choose the body before listing forces; each listed force must name an external agent acting on that body.
- State whether the observer is accelerating before deciding whether a pseudo-force is part of the chosen description.

Outside this book

- Momentum-conservation and collision algebra unless a verified syllabus or real question explicitly demands it.

- Routine non-inertial-frame or pseudo-force treatment as ordinary Grade-9 Standard syllabus content.
- Pulley-count formula catalogues, movable/compound pulley taxonomies, massive or elastic-rope models, and wedge-on-wedge systems.
- Variable-mass rocket dynamics, torque and rotational dynamics.

Friction opposes relative sliding at a contact

Subject: Physics

Topic: Laws of Motion

Concept: Friction opposes relative sliding at a contact

Concept difficulty: Medium concept

1

Before giving friction a direction, imagine the contact made perfectly slippery and ask which way the surfaces would slide relative to each other; friction opposes that, and is zero if there is no such tendency.

Medium concept

The key step

Friction is a contact force along the surface that opposes relative sliding, or the tendency for relative sliding, between the contacting surfaces; it is not defined by the object's overall velocity alone.

You need

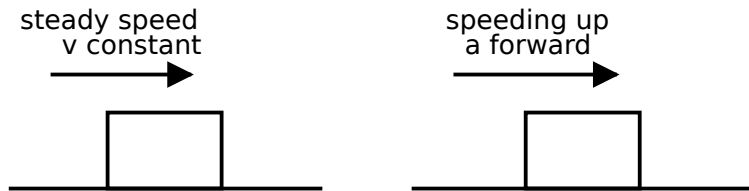
- Can choose one body and identify external contact interactions acting on it.
- Assign each force arrow to the body on which that force acts.

This unit builds toward

- **INJSO|2023|28 Jan 2023|Physics|Q24 · D2**

The move learners miss: A tyre is part of the chosen system, so a force attributed to the tyre is not the external agent that slows the entire rider+cycle system; the ground is. [Try it in Core2](#)

Book never slips on the tray



- 1 The step the hard question turns on

Choose the body or system and name the exact contact before deciding friction; for a contact between two bodies, keep the two friction forces as an equal-and-opposite third-law pair on different

Show the tray carried at steady speed and then speeding up.

Stage 1 of 3

Show the book on the tray in its two stages so friction's direction is found from the slip tendency, including the zero case.

bodies.

Why valid: Friction belongs to one contact interaction. The force on A by B and the force on B by A are a third-law pair, so neither may be moved onto the other body's free-body diagram.

Result: chosen body/system; named contact and partner; tangent direction; friction owner and receiver fixed.

2 Imagine the relative sliding, or tendency to slide, at that named contact if tangential friction were absent; point friction on the chosen body opposite that relative tendency.

Why valid: Contact friction resists relative sliding between

the surfaces.

Result: contact-specific slip tendency $\rightarrow$ friction direction; direction need not oppose the body's velocity.

- 3 Check a conveyor-belt case where the belt can frictionally accelerate an object in the same direction as the object's motion.**

Why valid: The counterexample falsifies the shortcut 'friction always points opposite velocity'.

Result: friction direction follows contact slip tendency, not a universal page-direction rule.

- 4 Decide whether any tangential friction is required at all; allow friction to be**

zero when the contact has neither relative sliding nor a tendency to slide.

Why valid: Friction is a responsive tangential contact force, not an automatic nonzero force merely because two rough surfaces touch.

Result: contact-state check -
> friction may be zero when no tangential force is required.

Equations and validity

Equation	What it tells you	When
$ f_s \leq \mu_s N$	For a contact that remains static, the static-friction magnitude adjusts to the value required by the tangential dynamics up to the	• Tr co re

Watch one

A book rests on a tray. A waiter first carries the tray at steady speed in a straight line, then speeds up forward; the book never slips. Find the friction force of the tray on the book in each

Equation	What it tells you	When
	limiting value $\mu_s N$; equality applies only at impending slip.	• N • St • Ec

stage, including whether it can be zero.

1. Choose the book as the body; the contact is the tray's top surface, and friction acts along it.
2. Steady carry: if the tray became perfectly slippery, the book would keep moving with the tray, so there is no slip tendency and the friction is zero.
3. Speeding up: without friction the book would keep its old speed while the tray moved ahead, so the book would slip backward relative to the tray.
4. Static friction opposes that tendency, so the tray's friction on the book points forward, the same way the book is moving.

Result

During the steady carry the friction is zero; while the tray speeds up, static friction on the book points forward.

Check

The book speeds up forward and friction is the only horizontal force on it, so by $F_{\text{net}} = m a$ that force must point forward, which agrees.

A tempting wrong path

- Friction always points opposite the object's velocity.
- Two rough surfaces in contact must always exert a nonzero friction force.

Diagnose

- A moving conveyor belt speeds up a box resting on it. Can the friction on the box point in the same direction as the box's motion?

1-2-3 QUICK CHECK

1 • CHECK

CHECK: For a chosen body, name the other surface at the contact and state which way that surface would slip relative to the body if friction vanished.

2 • APPLY

- A crate and rough platform already move together at constant velocity. If no horizontal force is otherwise required, must the friction force be nonzero merely because the surfaces are rough?

Repair

- Decide the relative sliding tendency at the contact first; friction opposes that tendency.
- No. Ask what relative sliding or slip tendency would occur without friction. If none exists and no tangential force is required, friction can be zero.

Try it with less support

A box is placed on a conveyor belt moving to the right faster than the box. Before the box matches the belt speed, what direction is the friction force on the box?

APPLY: A conveyor belt moves right faster than a box on it. Predict the friction on the box and the equal-and-opposite friction on the belt.

3 • CONNECT

CONNECT: Carry this contact ownership into the quantitative test: after direction is fixed, the next question is how much friction the no-slip motion requires and whether the contact can supply it.

Practice this concept in Core2

INJSO · 2023 · 28 Jan 2023 · Q24

Quantitative static and kinetic friction conditions

Subject: Physics

Topic: Laws of Motion

Concept: Quantitative static and kinetic friction conditions

Concept difficulty: Hard concept

1

Find the actual normal force and the friction that rest would need, compare that need with $\mu_s N$, and use μN as the friction value only at impending slip or once sliding.

Hard concept

Section 2 of 2

← Previous

The key step

Static friction is a responsive contact force: solve for the friction required to prevent relative slip and compare it with the limiting value $\mu_s N$. Equality is a threshold condition, while the kinetic model is used only after sliding is established.

You need

- Can determine friction direction from relative slip tendency.
- Can write Newton-II equations for a chosen body.

This unit builds toward

- **JEE (Advanced)|2020|Paper 1|Physics|Q13** · D4 · the hardest question in this set
The move learners miss: At the

- [Identify friction as a contact force that opposes relative sliding or the tendency to slide.](#)
- Relate signed net external force to acceleration with $F_{\text{net}} = m a$.

second stick-slip transition use kinetic friction at the slipping right contact and limiting static friction at the sticking left contact:

$\mu_k N_R = \mu_s N_L$; this gives $r = 50 = 25.6 \text{ cm}$ and swaps which finger slips. [Try it in Core2](#)

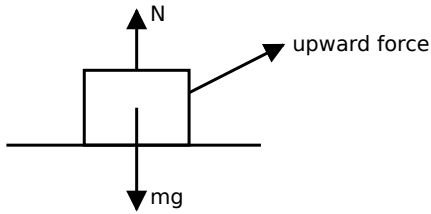
- **JEE (Advanced)|2014|Paper 2|Physics|Q19** · D3

The move learners miss: Compare the whole-system demand with m_2 's contact capacity: static rest requires $(m_1 + m_2)\tan\theta \leq 0.3m_2$, hence $\tan\theta \leq 0.2$. [Try it in Core2](#)

- **INJSO|2018|27 Jan 2018|Physics|Q12** · D3

The move learners miss: Use the three-person trial: $3F - f_k = ma = 50 \text{ N}$, then subtract the uniform-speed equation to isolate one person's force. [Try it in Core2](#)

1 · Solve perpendicular dynamics first



Use $\Sigma F_{\perp} = ma_{\perp}$. $N = mg$ only when justified.

If the equation requires $N < 0$: contact is lost.

CONTACT LOST $\rightarrow N = 0$ and $f = 0$

Choose the body and solve the perpendicular dynamics before touching a friction formula.

Stage 1 of 4

Build the complete dry-friction decision from actual normal reaction through static demand, impending slip, kinetic sliding and contact loss.

- 1 Build the chosen-body free-body diagram and solve the perpendicular dynamics for N before using any friction law: do not assume $N = mg$; $N=mg$ only in the special level case with no other perpendicular force or acceleration; a downward added component raises N , an upward component lowers it, and if the required contact force would be negative the contact is lost so $N=0$ and friction=0.**

Why valid: A unilateral contact can push but not pull. Its normal reaction is whatever the perpendicular Newton-II equation requires while contact persists;

friction capacity is then tied to that actual N .

Result: perpendicular equation $\rightarrow$ actual $N \geq 0$; level examples $N = mg + F_{\text{down}}$ or $N = mg - F_{\text{up}}$; contact lost $\rightarrow N = 0$ and $f = 0$.

- 2 **Assume no relative sliding provisionally, choose a body or a whole connected system deliberately, and solve the tangential Newton-II equation for the friction demand at each rough contact.**

Why valid: Static friction responds to the no-slip dynamics. A system boundary can remove internal contact pairs, but it cannot create friction

capacity at a smooth contact or move a rough contact from one body to another.

Result: chosen boundary + no-slip dynamics $\rightarrow$ signed f_{required} at each actual rough contact.

- 3 Compare each $|f_{\text{required}}|$ with $\mu_s N$; on an incline this comparison can be turned into the angle condition for the onset of sliding rather than setting $f = \mu_s N$ from the start.**

Why valid: The static-contact model permits any required value only up to the limiting magnitude.

Result: $|f_{\text{required}}| \leq \mu_s N \rightarrow$ static feasible; equality only

at impending slip; after sliding use the kinetic model. Equivalent symbolic form: $|f_{\text{required}}| \leq \mu_s N$.

4

The step the hard question turns on

At the threshold use $|f_s| = \mu_s N$; once relative sliding is established use $|f_k| = \mu_k N$ opposite the sliding, and if several contacts change state, update N and the friction model at each contact before continuing.

Why valid: Static-below-limit, impending-slip, kinetic-sliding and contact-lost are different contact states. A multi-contact process may switch which

contact is static or sliding, so state μN values cannot be carried across a transition.

Result: state map: static below limit $\rightarrow f_{\text{required}}$; impending $\rightarrow \mu_s N$; sliding $\rightarrow \mu_k N$; contact lost $\rightarrow N=f=0$.

Equations and validity

Equation	What it tells you	When
$ f_s \leq \mu_s N$	For a contact that remains static, the static-friction magnitude adjusts to the value required by the tangential dynamics up to the limiting value $\mu_s N$; equality applies only at impending slip.	• The contact remains static relative to the body. • Not applicable if the body is sliding.

Watch one

JEE (Advanced) · 2020 · Paper 1 · Q13 · Official past paper, faithful restatement of the original

A uniform metre scale is supported horizontally on two index fingers initially at 0.00 cm and 90.00 cm. Both fingers are moved slowly toward the centre. The slipping finger alternates because static and kinetic friction differ. With $\mu_s=0.40$ and

Equation	What it tells you	When
		as mg • Sta op im slip • Eq N i im
$ f_k = \mu_k N$	Once relative sliding at the contact is established, the bounded dry-sliding model uses kinetic-friction magnitude $\mu_k N$, directed opposite the relative sliding.	• Re be su est • N i the dy

$\mu_k=0.32$, find the final distance x_R of the right finger from the 50.00 cm centre at the instant it stops and the left finger starts slipping again.

1. **Treat the slowly moved metre scale as quasistatic and label the two finger supports by their current coordinates and normal reactions N_L and N_R .**

Why valid: At every instant the scale has negligible translational and angular acceleration, but the support reactions change as the finger positions change.

Result: $N_L+N_R=mg$; their ratio is set by support positions.

2. **For each current pair of support positions, obtain N_L and N_R from force and**

Equation	What it tells you	When
		• Th op slic coi • Th slic mc

torque balance before writing friction.

Why valid: Friction capacities are μN , so a fixed normal reaction while a support moves would make the threshold condition inconsistent with the changing support geometry.

Result: With the right finger at 90 cm, $N_L/N_R=40/(50-l)$; the first switch gives $l=18$ cm.

3. After the first switch, hold the left finger at 18 cm while the right finger slips inward and recompute the support ratio $N_R/N_L=32/(r-50)$.

Why valid: The support geometry has changed, so the normal reactions—and therefore both friction capacities—must be

recalculated for the second stage rather than carried over.

Result: $N_R/N_L = 32/(r-50)$.

4. **At the second stick-slip transition use kinetic friction at the slipping right contact and limiting static friction at the sticking left contact: $\mu_k N_R = \mu_s N_L$; this gives $r-50 = 25.6$ cm and swaps which finger slips.**

Why valid: The friction law is contact-state specific: a sliding contact uses μ_k while a sticking contact can rise only to μ_s . Equality marks the instant the roles switch.

Result: $x_R = 25.6$ cm.

Result

25.6 cm.

Check

The result is less than the initial 40 cm offset of the right finger and greater than zero, consistent with inward motion.

A tempting wrong path

- Friction always equals μN .
- The normal force is always mg .
- If both μ_s and μ_k are given, the coefficients themselves tell you which friction model to use.
- Friction can keep acting after the surfaces lose contact, or a negative value of N can be used inside μN .

Diagnose

- A small horizontal force is applied to a block that remains at rest. Must static friction already equal μ_s

1-2-3 QUICK CHECK

1 • CHECK

CHECK: State the four contact states in order and the friction rule for each: static below its limit, impending slip, established sliding, and contact lost.

2 • APPLY

APPLY: A level block has weight 50 N and is pulled 20 N upward as well as

N?

- If an extra force presses a block into a horizontal surface, can the normal reaction still be assumed equal to mg ?
- A problem gives both μ_s and μ_k but does not say whether the contact slips. Can you choose kinetic friction just because μ_k is supplied?
- An upward pull becomes large enough that the floor would need a downward normal force to keep touching the block. What are N and friction after separation?

Repair

- No. First solve the force required for no slip. Static friction matches that requirement up to its limit; equality occurs only at impending slip.
- Determine N from the perpendicular free-body equation. $N = mg$ is only a special case.
- No. Determine the contact state first: test whether the no-slip friction demand fits within $\mu_s N$. Use the kinetic model only after sliding is established.
- A contact can push, not pull. Once the perpendicular equation would require $N < 0$, contact is lost: set $N = 0$

12 N sideways. With $\mu_s = 0.5$, find N , the static limit, and the actual friction if it stays at rest.

3 · CONNECT

CONNECT: In a two-contact problem, after one contact starts sliding, recompute the normal reactions and identify which contact is now static before writing the next μN equation.

and therefore contact friction is zero.

Try it with less support

A contact has known normal reaction N and coefficient μ_s . The no-slip force equation requires tangential friction f_{required} . State the test for whether static contact is possible, when equality with $\mu_s N$ is justified, and what changes once sliding is established.

Practice this concept in Core2

IIT-JEE · 2007 · Paper I · Q10

IIT-JEE · 2008 · Paper II · Q33

IIT-JEE · 2011 · Paper I · Q41

JEE (Advanced) · 2014 · Paper 2 · Q19

JEE (Advanced) · 2020 · Paper 1 · Q13

JEE Main · 2026 · Session 2 · 06 Apr
2026 · Shift 2 · Q46

INJSO · 2011 · 29 Jan 2011 · Q22

INJSO · 2016 · 30 Jan 2016 · Q32A

INJSO · 2018 · 27 Jan 2018 · Q12

Physics · Force and laws of motion